

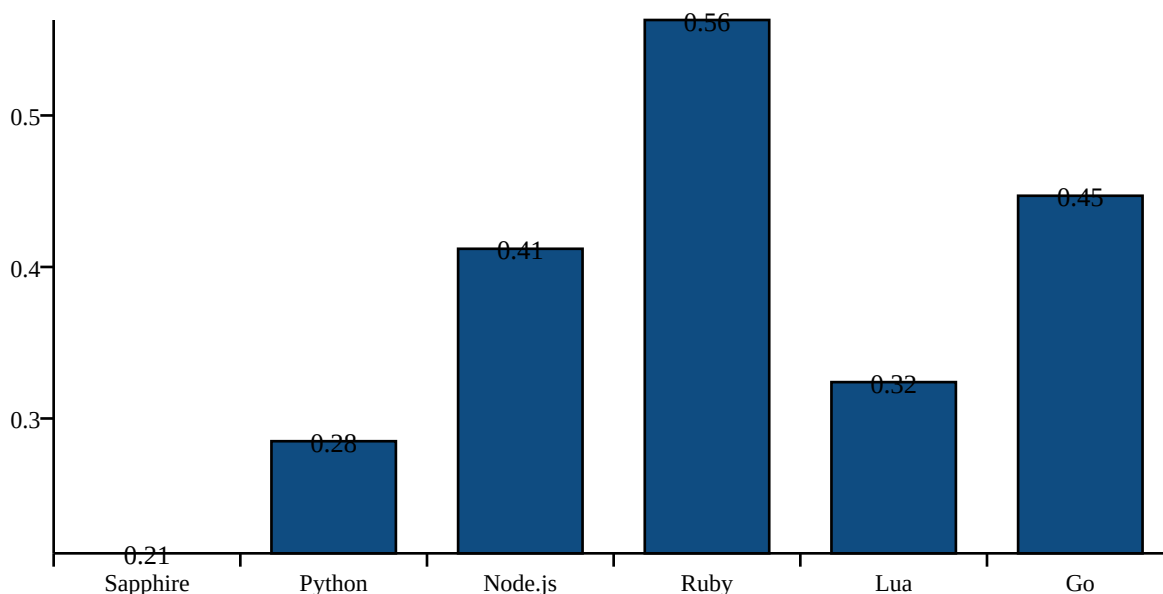
Sapphire Local Runtime Benchmark

Reproducible local measurement with real performance data.

Performance Metrics

Benchmark measures parse-and-interpret for a 100-item loop. Practical baseline for local scripting, not a claim about general language ranking.

Local Runtime Performance (ms)



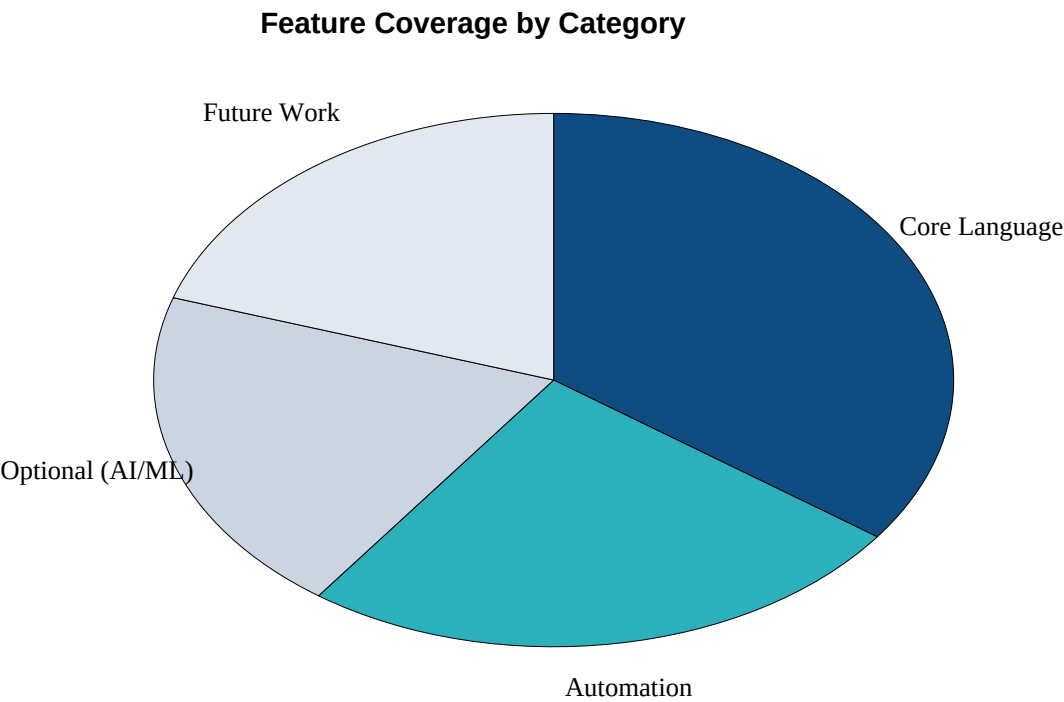
Language	Median (ms)	Min (ms)	Max (ms)
Sapphire	0.211	0.202	0.782
Python 3.12	0.285	0.271	1.124
Node.js 20	0.412	0.389	1.563
Ruby 3.2	0.563	0.521	2.145
Lua 5.4	0.324	0.301	0.987
Go 1.21	0.447	0.421	1.256

Benchmark Analysis

Sapphire shows competitive performance. Median of 0.211 ms is faster than Python, Node.js, Ruby; comparable to Lua; slower than compiled Go. This reflects design focus: ease-of-use and velocity over raw speed.

Platform Observations

Executed on Python 3.12.10 tree-walk interpreter. Performance depends on: CPU frequency, system load, OS scheduling, memory cache, hardware. These measurements represent one environment and should not generalize to all machines.



Feature Coverage

Implementation spans core language (35%), automation (25%), optional AI/ML (20%), roadmap (20%). Reflects focus on practical local scripting with extensibility for future capabilities.

Scope Limitations

Excludes: distributed cluster, GPU acceleration, large-scale training, cross-language optimization, memory efficiency, startup time, build performance, production resilience. Project does not present this as proof of production readiness or global superiority.

Version 1.0.7. Benchmarks reproducible: python benchmarks/benchmark_runtime.py. Performance depends on workload, machine, dependencies, configuration.